Where forced defects go: fractional corner charges, images and screening in dimer coverings

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When a region cannot be covered completely by dimers, some sites must stay empty, and their position is decided by entropy alone. The height field of a dimer covering is known to jump at the corners of a region. We treat these jumps as fixed fractional charges $q_c = -J_c/z$ acting on the mobile defect, which is a vortex of winding z, with straight boundaries acting as mirrors of the same sign and the stiffness g setting the coupling. This gives one exponent for the defect density near a fixed charge at a corner, on an edge or in the interior, a rule for which representative of J_c modulo z dominates, and a criterion for binding. We test the picture with exact Pfaffian calculations and Monte Carlo simulations. (i) On the square lattice the image doubles the coupling, so a fixed charge that leaves the defect free in the interior binds it at an edge. With interacting dimers at $g \approx 1.9$, a single vacancy behaves likewise: at an edge the local decay exponent of the near-vacancy probability falls toward zero, in the interior it does not. (ii) With interacting dimers the ratio of edge to bulk pair exponents stays at 2.01 ± 0.05 while g varies twentyfold, and the corner exponent follows $g/2$ within 4% for $g \leq 1$. (iii) In honeycomb regions with armchair edges the boundary-layer offsets of the height on the two classes of edges differ by $\frac{1}{2}$, so the charge at a 120° corner is shifted by $\pm\frac{1}{6}$ from the value set by the lattice jump. A rotation symmetry makes the offsets opposite, the Temperley property of one family of hexagons fixes their difference within the Coulomb-gas description, and an exact strip calculation gives 0.5001 ± 0.0010. With this shift the density formula reproduces full defect densities on hexagons to 3–5% with only the normalization fitted and predicts the corner exponents of regions that are not Temperleyan. (iv) Diagonal bonds of weight t break bipartiteness and screen the force beyond a length of order $1/t$. At the critical value $t = 1/2$ of Basor and Bleher, the pair correlation along non-anti-diagonal directions equals $[1 - (\sqrt{3} - 2)^d]/6$ to 40 digits, consistent with the Toeplitz symbol becoming rational there.

I. INTRODUCTION

Take a chessboard, remove one corner square, and cover the rest with dominoes [Fig. 1(a)]. The board now has one more white square than black, so one white square is always left uncovered. If all coverings are equally likely, where is that square? The answer is far from uniform [Fig. 1(b)]. The empty square avoids the corner opposite the cut and is drawn to the cut, and its density near every corner and edge follows a power law whose exponent depends on the corner. Nothing in the problem has energy; every bias comes from counting coverings.

The same question arises wherever a constrained medium must accommodate a defect it cannot remove. In resonance theory, the radical centre of a benzenoid molecule with one more site on one sublattice than on the other is weighted by the number of Kekulé structures of the remaining graph, which is exactly the dimer problem with one forced monomer. Vacancies and holons in short-range resonating-valence-bond states and quantum dimer models raise it too [1–3], as do holes in tilings [4–7] and the monomer-dimer problem itself [8–11]. In each case the defect feels an entropic force, and in a finite flake or sample that force is dominated by boundaries and corners.

The tool is the height description of close-packed dimers [3, 12–15]. The covering constraint defines a height function. Its long-wavelength fluctuations are those of a Gaussian free field (GFF) with stiffness g, and a site left uncovered is a vortex of that field [16, 17]. In the bulk, gaps in dimer coverings interact by two-dimensional electrostatics: Ciucu proved this for lozenge tilings [4–6], for collinear gaps on the square lattice [18], and for doubly periodic planar bipartite graphs in the liquid phase [7]. Near a boundary, the image of a gap keeps its sign at constrained boundaries and reverses it at free ones [19–21]. The boundary itself is also understood: the mean height jumps at corners of the region. Russkikh showed that on square-lattice domains the boundary height shifts at each corner according to its colour and convexity [22], Berestycki and Liu proved that the height converges to a GFF whose boundary values jump at every non-Temperleyan corner [23], and Allegra identified the jumps with boundary-condition-changing (bcc) operators [24]. Monomers on a straight boundary of the square lattice have exactly computable correlations [25].

What these results leave open is the force on a defect that is free to move. Here we read the corner jumps as fixed fractional charges $q_c = -J_c/z$ acting on that defect, where J_c is the jump at the corner and z is the winding of a defect. Straight edges act as same-sign mirrors, and the stiffness sets the coupling. For the square lattice at the free-fermion point this picture reproduces exact defect densities on rectangles and on boards with removed

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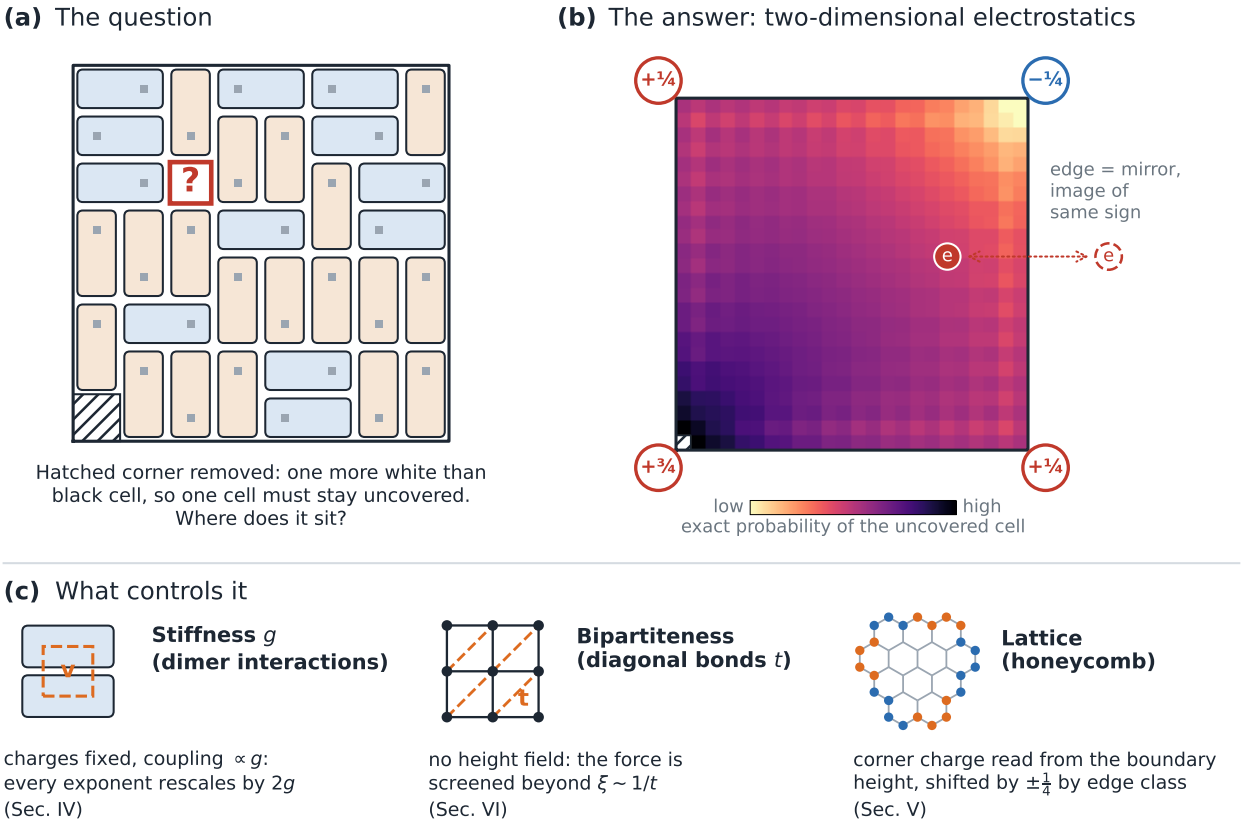


FIG. 1. The problem and its electrostatic answer. (a) An 8×8 board with one corner removed has one more white than black cell, so every dimer covering leaves one white cell uncovered (a sample covering; grey squares mark black cells). Where is that cell? (b) Exact probability of the uncovered cell on a 24×24 board with the same corner removed (dark: high). The cell is drawn to the removed corner and pushed away from the opposite one. This is what a charge $e = -1$ would do in the field of fixed charges at the corners ($q_c = +\frac{3}{4}$ at the removed corner, $\pm\frac{1}{4}$ at the others), with each edge acting as a mirror that makes an image of the same sign. (c) The three controls studied here.

corners to 1–3% [26]. Section II states it for a bipartite lattice of coordination z and stiffness g. It gives a single exponent formula for a fixed charge at a corner, on an edge, or in the interior, a rule for choosing among the representatives of J_c modulo z, and a criterion for binding. It also shows that in Temperleyan regions every corner exponent equals $g/2$, so such regions cannot discriminate between corner rules, but fix the corner charges exactly.

The remaining sections test the picture in the three directions of Fig. 1(c). Section III tests binding exactly on the square lattice: the same charge binds a defect at an edge but not in the interior. Section IV changes the stiffness with interacting dimers. Section V moves to honeycomb regions with armchair edges, where the corner charge at 120° corners needs one new ingredient, an edge-dependent boundary-layer offset of the height that we determine by symmetry, by the Temperley property and by an exact strip calculation. Section VI breaks bipartiteness and shows how the entropic force is screened. Section VII collects the results.

II. ELECTROSTATICS OF FORCED DEFECTS

A. Heights, defects and corner charges

Let the lattice be bipartite with sublattices black and white, and let every site have z neighbours ($z = 4$ on the square lattice, 3 on the honeycomb). Thurston's height [12] lives on the faces. Crossing an edge with a black site on the left, it changes by $+1$ if the edge is empty and by $-(z-1)$ if the edge carries a dimer. Around a covered site the changes add up to zero, so the height is single valued. Around an uncovered site they add up to $\pm z$, so a forced defect is a vortex of winding z. We give black defects the charge $e = +1$ and white defects $e = -1$, and call a region boundary *closed* when no dimer crosses it; this is a constrained boundary in Ciucu's terminology [20].

Along a closed boundary the height is fixed by the shape alone. Along a straight edge it oscillates on the lattice scale around a well-defined mean. At a corner the

mean jumps by an integer J_c, as found rigorously on the square lattice [22, 23]. The jumps around the boundary add up to z times the net defect charge.

The jumps act as charges. In the upper half plane, the harmonic function with boundary value 0 on $x<0$ and J on $x>0$ is $(J/\pi)\arg z$, a vortex of winding J over the half circle. A defect of winding z on the boundary, together with its same-sign image, produces the same function with J replaced by z, because the two together wind by z over the half circle. A jump J_c is therefore equivalent to a fixed charge

$$q_c = -\frac{J_c}{z} \tag{1}$$

at the corner, in units of the defect charge (the sign follows from orienting the boundary counterclockwise). The corner charges are fractional and neutralize the defects, $\sum_c q_c = -\sum_i e_i$. On the board of Fig. 1(b) the corners of the full board carry $\pm\frac{1}{4}$, and the removed corner carries $+\frac{3}{4}$.

The boundary steps fix every jump except for one choice: the total winding $z\sum_i e_i$ enters at the corner where the branch cut of the defect meets the boundary, and moving the cut to another corner changes two jumps by $+z$ and $-z$. The individual J_c are therefore defined modulo z, with their sum fixed. In the Coulomb gas the corner is a bcc operator whose dimension is proportional to $(\pi/\theta_c)J_c^2$ at a corner of interior angle θ_c [24, 27]. The leading behaviour is set by the lowest total dimension, so we fix the representatives once and for all by

$$\min_{\{J_c\}} \sum_c \frac{\pi}{\theta_c} J_c^2 \quad \text{subject to} \quad \sum_c J_c = z\sum_i e_i. \tag{2}$$

The same rule is applied to the effective jumps of Sec. V.

B. Density formula and exponents

We write the stiffness as g, normalized so that a pair of opposite defects in the bulk correlates as r^{-g}; $g=1/2$ for non-interacting dimers on the square and honeycomb lattices, and g is the constant of the Coulomb-gas analysis of interacting dimers by Alet *et al.* [16, 17]. A closed boundary is a Dirichlet boundary for the height (equivalently, Neumann for the dual field), which makes the images of vortices carry the same sign. Map the region conformally onto the upper half plane by $w=f(z)$, with the corners going to real points f_c. The Coulomb-gas prediction for the weight of a mobile defect of charge e at z is

$$\rho(z) \propto |f'(z)|^{g/2} (\operatorname{Im} f)^{g/2} \prod_c |f(z)-f_c|^{2g\,eq_c}. \tag{3}$$

The first factor is the self-energy of the vortex, the second its interaction with its own image, and the product its interaction with the corner charges. Boundary charges coincide with their images, which doubles their coupling.

TABLE I. Density exponent γ near a fixed charge q for a mobile defect e, $\rho \sim d^{\gamma}$. The defect is bound when $\gamma<-2$.

position	γ	bound if
corner, angle θ	$2g[(\pi/\theta)(\frac{1}{2}+eq)-\frac{1}{4}]$	$\gamma<-2$
straight edge	$2g(\frac{1}{4}+eq)$	$eq<-\frac{1}{g}-\frac{1}{4}$
interior	geq	$eq<-\frac{2}{g}$

A fixed charge q in the interior at f_q contributes $|f-f_q|^{geq}|f-\bar{f}_q|^{geq}$ instead. At $g=1/2$ Eq. (3) is the formula tested on rectangles in Ref. [26].

Near a corner of interior angle θ, $f-f_c \sim z^{\pi/\theta}$, so $\rho \sim d^{\gamma}$ at distance d from the corner with

$$\gamma = 2g\left[\frac{\pi}{\theta}\left(\frac{1}{2}+eq\right)-\frac{1}{4}\right]. \tag{4}$$

A charge on a straight edge is the case $\theta=\pi$, $\gamma=2g(\frac{1}{4}+eq)$, and a charge in the interior gives $\gamma=geq$ (Table I). Without charges these reduce to the known exponents $x_c=(\pi/\theta)x_s$ of corner operators [27]. On a region of linear size L the probability to find the defect within a fixed distance of the charge scales as $L^{-(2+\gamma)}$ for $\gamma>-2$, and the corner-to-centre ratio as $L^{-\gamma}$.

Three consequences follow. First, g enters Eq. (3) only through the overall power, $\rho_g \propto \rho_{1/2}^{2g}$, provided the corner charges do not depend on g. All exponents are then rescaled by the same factor. This holds whenever the boundary layer cannot shift the height, which on the square lattice is guaranteed by symmetry (Sec. V C). Second, the near-charge weight $\int d^{\gamma}\, d\,\mathrm{d}d$ converges when $\gamma<-2$, and the defect is then bound to the charge with a probability that stays finite as $L\to\infty$. The edge exponent contains $2eq$ while the interior one contains eq, because the image doubles the attraction. At $g=1/2$ a charge $eq=-3$ binds at an edge but not in the interior. A single vacancy ($eq=-1$) binds at an edge for $g>4/3$ but in the interior only for $g>2$.

Third, some regions fix the corner charges by themselves. In a Temperleyan region [15, 28] the defect probability is the same at every boundary site of the majority sublattice [10]. By Eq. (3), a uniform boundary density requires $|f'|^g\prod_c|f-f_c|^{2geq_c}$ to be constant along the boundary, and near each corner this gives

$$eq_c = -\frac{1}{2}\Big(1-\frac{\theta_c}{\pi}\Big), \qquad \gamma=\frac{g}{2} \quad \text{at every corner.} \tag{5}$$

This gives $eq_c=-\frac{1}{4}$ at 90°, $-\frac{1}{3}$ at 60° and $-\frac{1}{6}$ at 120°. Corner exponents measured in Temperleyan regions therefore cannot distinguish between rules for q_c; they test Eq. (3) as a whole, and they fix the charges. Only regions that are not Temperleyan discriminate.

The assumptions are that the boundary condition of the height stays Dirichlet for all g, that no locking operator is relevant (which fails near the columnar transition of interacting dimers [16]), and that the region is a flat liquid, i.e. its boundary does not force a nonzero tilt.

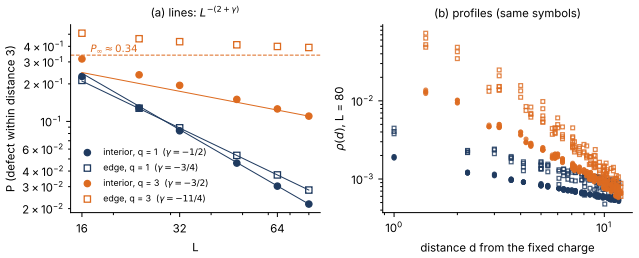


FIG. 2. A fixed charge in the interior and at an edge, square lattice, exact. (a) Probability that the mobile defect is within distance 3 of the charge, against board size L; lines show $L^{-(2+\gamma)}$. The edge charge $q = 3$ is bound; the dashed line is the extrapolated P_∞. (b) Density profile against distance from the charge at $L = 80$.

TABLE II. Fixed charges on the square lattice ($g = 1/2$, $e = -1$), $L = 16$–80: predicted and measured scaling of $P \propto L^{-(2+\gamma)}$ (range of local slopes, smallest to largest L) and of the profile $\rho \propto d^{\gamma}$.

charge	γ	P: pred.	local slopes	profile
interior $q = 1$	$-\frac{1}{2}$	-1.50	$-1.44 \to -1.48$	-0.50
edge $q = 1$	$-\frac{3}{4}$	-1.25	$-1.27 \to -1.25$	-0.68
interior $q = 3$	$-\frac{3}{2}$	-0.50	$-0.72 \to -0.60$	-1.47
edge $q = 3$	$-\frac{11}{4}$	0 (bound)	$-0.25 \to -0.09$	-2.3

III. FIXED CHARGES AT AN EDGE AND IN THE INTERIOR

We first test binding where everything is exact: the square lattice at $g = 1/2$. On an even $L \times L$ board, a fixed charge is made by removing cells. One removed black cell is a charge $q = 1$ and leaves one mobile white defect, $e = -1$. A plus-shaped hole (four black cells around a white one) is $q = 3$; two further white cells, removed at fixed fractions of L away from it (Appendix A), restore the balance and leave one mobile white defect. We place the fixed charge at the centre or on the boundary, compute the exact defect probability at every site with Kasteleyn matrices, and record the probability P that the defect is within Euclidean distance 3 of the centre of the charge (Fig. 2).

Table II shows the outcome. For $q = 1$ the local slopes approach the predictions from both sides. For the interior $q = 3$ the local slope drifts toward $-1/2$; the profile slope -1.47 agrees with $-3/2$, and the defect is not bound. The same hole on an edge binds it. $P = 0.51, 0.46, 0.44, 0.41, 0.40, 0.39$ for $L = 16$–80, and the local slope falls monotonically, $-0.25, -0.19, -0.14, -0.11, -0.09$. A power law with a small exponent would keep a constant local slope; a bound defect, $P = P_\infty + bL^{2+\gamma}$ with $L^{2+\gamma} = L^{-3/4}$, drives it to zero, and a fit of that form to $L \geq 24$ gives $P_\infty \approx 0.34$. Equation (4) decides whether the defect is bound; it does not predict P_∞, which is set on the lattice scale. With $\gamma = -11/4$ most of the edge weight sits within a few lattice spacings, which is why the profile slope (-2.3) is not yet at its continuum value. The image doubles the coupling, and the doubling turns a delocalized defect into a bound one.

IV. CHANGING THE STIFFNESS

Interactions between dimers change g without changing the winding of the defects. We use the model of Alet *et al.* [16, 17], with weight $e^{-\beta v N_\parallel}$, where $N_\parallel$ is the number of plaquettes holding two parallel dimers. Aligning interactions ($\beta v < 0$) raise g toward the columnar transition at $\beta v \approx -1.5$, and repulsive ones lower it. On the square lattice the corner charges stay fixed for every g (Sec. V C), so Eq. (3) predicts that every exponent is multiplied by $2g$. We test three consequences that have no free parameter (Fig. 3; methods and error analysis in Appendix B).

Pair exponents. A defect pair in the bulk decays as d^{-g}; along a straight edge it decays as d^{-2g}, because each defect also interacts with the other's image. The ratio is 2 for every g; at $g = 1/2$ both exponents are known exactly [9, 25], and the doubling is the standard behaviour of boundary operators of a compact boson. We sampled pair weights with a worm algorithm on a cylinder of circumference 64 and height 128, with the tail in the bulk or on the open edge. For $\beta v = +1, +0.5, 0, -0.4, -0.7, -1$ the bulk exponent is $g = 0.092$, 0.243, 0.506, 0.839, 1.224 and 1.880 (exact 1/2 at $\beta v = 0$), fitted over chord distances 4–20. The edge-to-bulk ratio is 2.11, 1.99, 1.95, 1.98, 2.01 and 2.00; the six values have mean 2.01 and standard deviation 0.05 [Fig. 3(a,b)]. The images keep their sign at all g.

Corner exponent. On an odd $L \times L$ board one black defect is forced, and each corner carries $q_c = -\frac{1}{4}$. Equation (4) with $\theta = \pi/2$ gives $\gamma = g/2$; the odd board is Temperleyan at $g = 1/2$, so this value is the one of Eq. (5). At $g \neq 1/2$ the D_4 symmetry still fixes $J_c = 1$ at every corner, and the mirror argument of Sec. V C excludes an edge offset, so $\gamma = g/2$ tests Eq. (3) at general g. We measure the corner-to-centre ratio, averaged over black sites within Manhattan distance 2 of the corners and of the centre, for $L = 17$–65 [Fig. 3(c)]. Fitting $L = 25$–65 gives $\gamma = 0.116 \pm 0.005$, 0.253 ± 0.006, 0.464 ± 0.008 and 0.853 ± 0.023 for $\beta v = +0.5$, 0, -0.5 and -1, against $g/2 = 0.121$, 0.253, 0.484 and 0.940 (g at $\beta v = -0.5$ interpolated between the cylinder values). To separate the estimator from the physics, we computed the same averaged ratio exactly at $\beta v = 0$ with Kasteleyn matrices: it gives 0.243 over $L = 25$–65, with local values rising to 0.247 at $L = 129$, so the estimator approaches $g/2$ slowly from below; the Monte Carlo value 0.253 ± 0.006 agrees with it within 1.7σ. For $g \leq 1$ the corner exponent is within 4% of $g/2$ (at $\beta v = -0.5$, where g is interpolated, 2.5σ). At $g = 1.88$ it is 9% (3.8σ) be-

low; the vacancy test below shows the same slow convergence at this coupling, which is within a factor 1.6 of the columnar transition in βv.

Binding of a vacancy. The sharpest prediction of Table I is that a single removed site ($eq = -1$) binds the mobile defect at an edge when $g > 4/3$ but in the interior only when $g > 2$. At $\beta v = -1$ ($g = 1.88$) the edge vacancy should therefore bind ($\gamma = -2.82$) while the interior vacancy should not ($\gamma = -1.88$). At $\beta v = -0.4$ ($g = 0.84$) neither should. We removed one black site at the centre or at the midpoint of the bottom edge of an even $L \times L$ board, $L = 16$–64, and measured the probability P that the white defect is within distance 3 of it [Fig. 3(d)]. At $\beta v = -0.4$ both P decay: the local slopes are $-1.13 \to -1.18$ in the interior (predicted $-(2+\gamma) = -1.16$) and $-0.87 \to -0.82$ at the edge (predicted -0.74). At $\beta v = -1$ the two positions behave differently. At the edge the local slope falls monotonically, $-0.23, -0.18, -0.12, -0.10$, as for the bound $q = 3$ charge of Sec. III, and P levels off near 0.5. In the interior it stays at -0.49 ± 0.02 for all sizes, so the defect is not bound. The interior slope is, however, steeper than the asymptotic -0.12: the ratio of the weight at distances 6–12 to that within 3 still grows from 0.60 to 0.70 between $L = 32$ and 64, so the interior profile has not converged at these sizes, and we do not claim a quantitative interior exponent at this coupling. The qualitative prediction, binding at an edge but not in the interior for $4/3 < g < 2$, is borne out.

V. THE HONEYCOMB LATTICE

A. Flat regions and forced defects

Honeycomb dimers are lozenge tilings. A defect winds the height by 3, so Eq. (1) becomes $q_c = -J_c/3$, and $g = 1/2$. Corners now come at two angles, 60° and 120°.

Equation (3) needs regions whose interior is a flat liquid, which on the honeycomb means armchair boundaries. Zig-zag regions are lozenge hexagons, which freeze near their corners into the arctic circle [30]. Armchair flakes stay flat: the local deviation of the bond probabilities from 1/3 decreases with size, and no bond is nearly frozen (Fig. 10). Armchair hexagons, triangles and parallelograms centred on an A site have one more A than B site, so one A defect ($e = +1$) is forced without removing anything (Fig. 4). Along an armchair edge the boundary steps of the height alternate between $+1$ and -1, so the mean height is constant and jumps only at corners (Fig. 11). Hexagons fall into two families: type I with $J = 0$ and $+1$ on alternating corners, and type II with $J = -1$ and $+2$. Both satisfy Eq. (2) and $\sum_c J_c = 3$.

B. Temperleyan regions fix the corner charges

Type I hexagons are Temperleyan in the sense of Sec. II: the defect probability is the same at every outermost A site to 10^{-13} (for $R = 28$ and 36), and the whole density is sixfold symmetric to 10^{-16} although the region itself has only threefold symmetry (Fig. 12). Type II hexagons are not (boundary probabilities vary by a factor 6–9). By Eq. (5), all six corners of a type I hexagon must carry $eq_c = -\frac{1}{6}$. The lattice jumps $J = 0$ and $+1$ would instead give 0 and $-\frac{1}{3}$. At 120° corners the effective jump is therefore $J \pm \frac{1}{2}$, not J. At 60° corners Eq. (5) gives $-\frac{1}{3}$, which is what the lattice jump $J = +1$ of armchair triangles gives; here the lattice jump is already right.

The measured corner exponents of all Temperleyan regions, which must equal $\frac{1}{4}$ exactly, come out as 0.28 ± 0.02 (type I hexagons and the F2 parallelograms below). This $+0.03$ is the bias of fitting corner-to-centre ratios at these sizes ($R \le 45$); the same fit gives 0.26 ± 0.11 for triangles.

C. The edge-class offset

The value of the height that enters the continuum theory is not the lattice boundary height but the value to which the interior field extrapolates at the edge. The two can differ by an offset δ set on the lattice scale. Two symmetries constrain it.

On the square lattice, a mirror perpendicular to a straight edge and passing through cell centres maps the edge to itself, preserves the colours and reverses orientation, so it maps $h \to c - h$. Applied to the offset, which does not depend on position along the edge, it gives $\delta = -\delta$, i.e. $\delta = 0$. The argument uses only lattice symmetries, so it holds for every g, with or without interactions; on the square lattice the corner charges are the lattice jumps for all g.

On the honeycomb lattice every mirror perpendicular to an armchair edge exchanges the two sublattices as well as reversing orientation, so it leaves h invariant and imposes nothing. Armchair edges come in two classes by the direction of their outward normal (Fig. 4). A 180° rotation about the centre of a hexagonal face exchanges the two classes and the two sublattices, and preserves orientation, so it maps $h \to c - h$ and

$$\delta_- = -\delta_+. \tag{6}$$

A 60° corner joins two edges of the same class and is unaffected. A 120° corner joins edges of different classes, so its effective jump is $J \pm 2\delta$. Symmetry fixes neither the size nor the sign of δ_+. The Temperley property fixes both: matching $J \pm 2\delta$ to $J \pm \frac{1}{2}$ above gives $2\delta = \frac{1}{2}$ exactly, within the Coulomb-gas description. An independent exact calculation on an infinite armchair strip of half-width R (Appendix A) gives $\delta_- - \delta_+ = 0.402$, 0.484, 0.484, 0.489, 0.505, 0.500, 0.502, 0.499, 0.4995 and

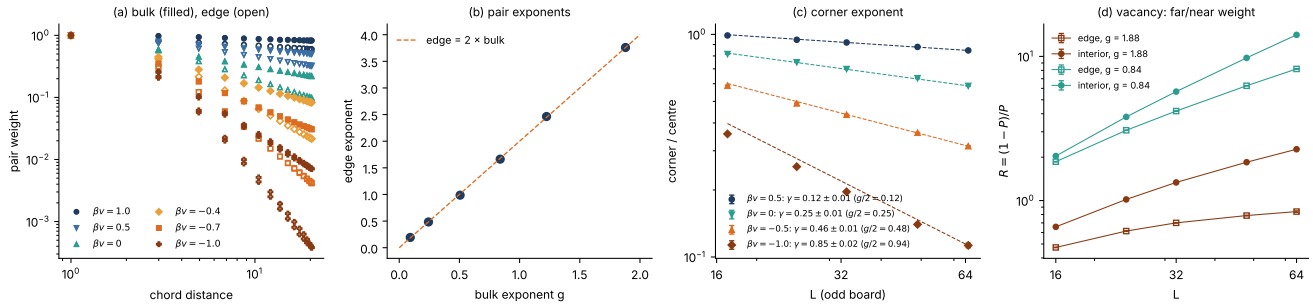


FIG. 3. Interacting dimers, Monte Carlo. (a) Pair weight against chord distance on a cylinder, tail in the bulk (filled) or on the open edge (open). (b) Edge exponent against bulk exponent; the dashed line is $2g$. (c) Corner-to-centre ratio of a forced defect on odd boards against L, with lines of slope $-g/2$. (d) Vacancy binding: $R = (1-P)/P$, the weight away from a removed site relative to the weight within distance 3, for an interior and an edge vacancy. A bound defect has $R \to$ const. Error bars are from ten blocks.

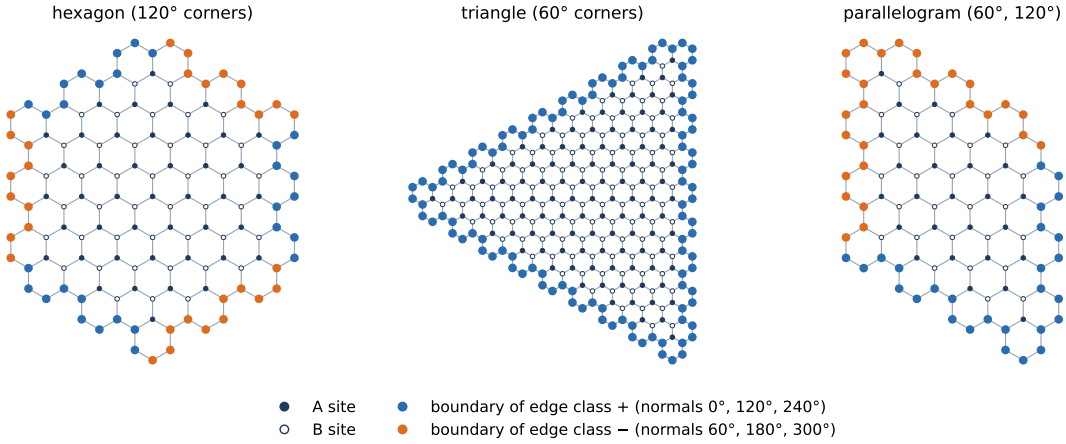


FIG. 4. Armchair regions on the honeycomb lattice, with boundary sites coloured by the class of their edge (outward normals $0°, 120°, 240°$: class +; $60°, 180°, 300°$: class −). Hexagons have only 120° corners, each joining edges of different classes; triangles have only 60° corners, joining edges of the same class; parallelograms have both. Corners are named below by the polar angle at which they sit (e.g. the 30° corner).

0.5003 for $R = 12$, 20, 30, 40, 60, 80, 120, 160, 240 and 320 [Fig. 5(b)]. For $R \geq 80$ the values scatter around 0.5001 with standard deviation 0.0010 and no trend in R; the scatter is larger than the spread between quadrature grids and comes from the choice of the window in which the interior profile is fitted. This strip value is the independent determination; the Temperley argument relies on Eq. (3) itself, and also assumes that the lattice-scale amplitude of the boundary density is the same on both classes of edges. The corner charge is

$$q_c = -\frac{J_c + \sigma_c/2}{3}, \tag{7}$$

where $\sigma_c = +1$ (-1) if, going counterclockwise, the corner leads from a class + (class −) edge to a class − (class +) edge, and $\sigma_c = 0$ at 60° corners; the representatives are chosen by Eq. (2) applied to $J_c + \sigma_c/2$. A convention-free statement is that the charge at a 120° armchair corner is a half-odd integer divided by 3. In the language of bcc operators [24], the offset changes the operator at a 120° corner from the one set by J to the one set by $J \pm \frac{1}{2}$. Whether a further offset localized at the corner itself exists is not excluded by symmetry; the corner exponents below show no sign of one.

D. Tests in regions that are not Temperleyan

Type II hexagons and armchair parallelograms are not Temperleyan, so their corner exponents test Eq. (7). For type II, Eq. (7) gives $q_c = +\frac{1}{6}$ and $-\frac{1}{2}$ at the two kinds of corner, i.e. $\gamma = \frac{3}{4}$ and $-\frac{1}{4}$; we measure 0.75 ± 0.07 and -0.25 ± 0.02 (Fig. 13). The lattice jumps would give 1 and $-\frac{1}{2}$.

Parallelograms have two 60° and two 120° corners. Family F2 turns out to be Temperleyan: its four corner values agree to 10^{-10} although two corners are 60° and two 120°, and all four give 0.28 ± 0.02, as for type I. Fam-

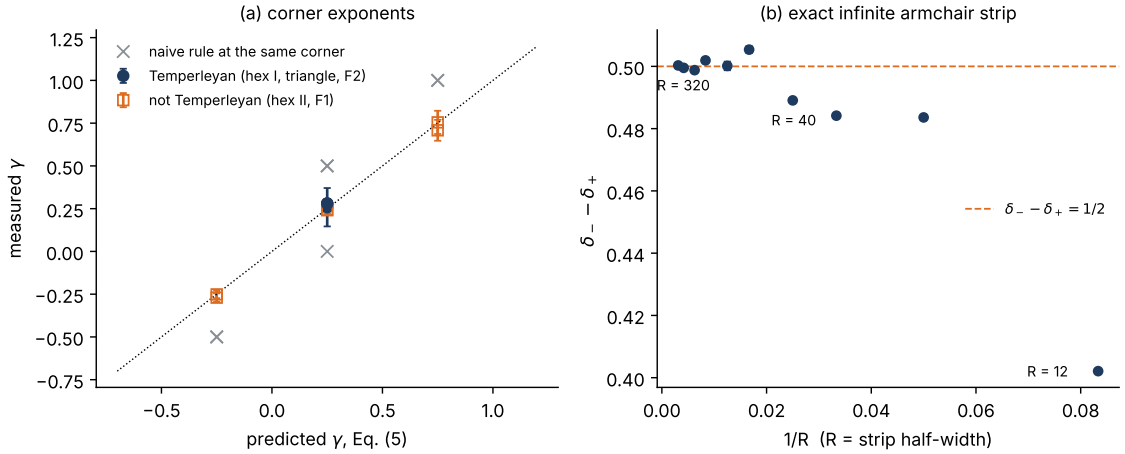


FIG. 5. The edge-class rule on the honeycomb lattice. (a) Measured against predicted corner exponents with fit errors. Filled circles: Temperleyan regions (type I hexagons, triangles, F2 parallelograms), whose exact value is $\frac{1}{4}$; open squares: regions that are not Temperleyan (type II hexagons, F1 parallelograms). Crosses: the naive rule $q_c = -J_c/3$ where it differs. (b) Edge-class offset $\delta_- - \delta_+$ from exact infinite strips against $1/R$; it tends to $\frac{1}{2}$.

TABLE III. Corner exponents of armchair parallelograms. Corners are named by their polar angle; θ is the interior angle. J is the lattice jump in the representative of Eq. (2). Errors are fit errors from the largest sizes; the Temperleyan family F2 shows the fit bias (+0.03).

family	corner (θ)	J	Eq. (7)	naive	measured
F1	30° (120°)	−1	+0.75	+1.00	$+0.71 \pm 0.06$
F1	120° (60°)	+1	+0.25	+0.25	$+0.24 \pm 0.02$
F1	210° (120°)	+2	−0.25	−0.50	-0.27 ± 0.03
F1	300° (60°)	+1	+0.25	+0.25	$+0.24 \pm 0.02$
F2	all four	—	+0.25	—	$+0.28 \pm 0.02$

ily F1 is not. Its exponents were predicted from Eq. (7) before they were computed (Table III). The two 120° corners discriminate, and they lie within 0.8σ of Eq. (7). The naive rule $q_c = -J_c/3$ is not even unique here: under Eq. (2) it is tied between $(J_{30}, J_{210}) = (-1, 2)$ and $(2, -1)$, which predict $(1, -\frac{1}{2})$ and $(-\frac{1}{2}, 1)$; even the better of the two lies 4.8σ and 9.2σ from the data. The two 60° corners of F1, where both rules give $\frac{1}{4}$ in a region that is not Temperleyan, agree as well.

E. Full densities

Exponents probe the density only near corners. Equation (3) predicts the whole density, which we compared with exact densities on hexagons of both types. The map to the upper half plane is a Schwarz–Christoffel map (Appendix A), the corner charges come from Eq. (7) or from the lattice jumps, and only the overall normalization is fitted. Figure 6 shows the profiles toward a 30° corner, a 90° corner and an edge midpoint. Excluding the outermost 3 (6) rows, Eq. (7) reproduces $\ln \rho$ with an rms deviation of 4.1–6.1% (3.2–4.6%) on type I hexagons with $R = 28, 36$ and type II with $R = 27, 38$; the naive rule deviates by 10–14% (7.3–9.7%), with systematic errors toward the corners. For type I this is a consistency check of Eq. (3), since the charges are fixed by the Temperley property; for type II it is a test. The result does not depend on whether the effective boundary is placed at the outermost sites or up to one lattice spacing outside.

F. Images on armchair edges

At the midpoint of an armchair edge, a same-sign image predicts a defect probability $\propto R^{-1/4}$ relative to the centre and a profile rising as $d^{+1/4}$ into the bulk; an opposite-sign image would give $R^{+1/4}$ and $d^{-1/4}$. We find $R^{-0.29}$ (type I) and $R^{-0.24}$ (type II), and inward slopes +0.20 and +0.21 over $2 < d < R/4$ (Fig. 14). This is consistent with Ciucu's rule that constrained boundaries keep the sign of images [20].

VI. BREAKING BIPARTITENESS

The height function exists only on bipartite lattices. To see what replaces the electrostatics without it, we add a bond of weight t across one diagonal of every square plaquette [Fig. 9]. At $t = 0$ this is the square lattice; at $t = 1$ it is the triangular lattice. The graph stays planar, so Pfaffians remain exact (Appendix A). In the bulk the defects deconfine for every $t > 0$ [2], the limit of the pair function is known in closed form [31], and its approach was found by Basor and Bleher [32]. We add the behaviour of a forced defect in a finite region and one

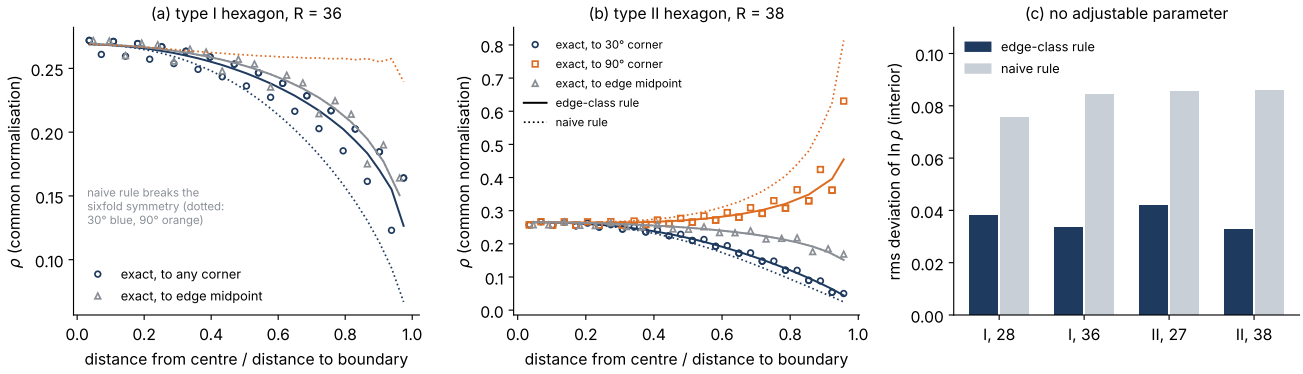


FIG. 6. Full defect density on armchair hexagons, exact (symbols) and Eq. (3) with no adjustable parameter (lines). (a) Type I, $R = 36$: the exact density is sixfold symmetric, so the profiles to the 30° and 90° corners coincide. (b) Type II, $R = 38$. Solid lines: Eq. (7); dotted lines: naive rule. (c) rms deviation of $\ln \rho$ excluding the outermost 6 rows.

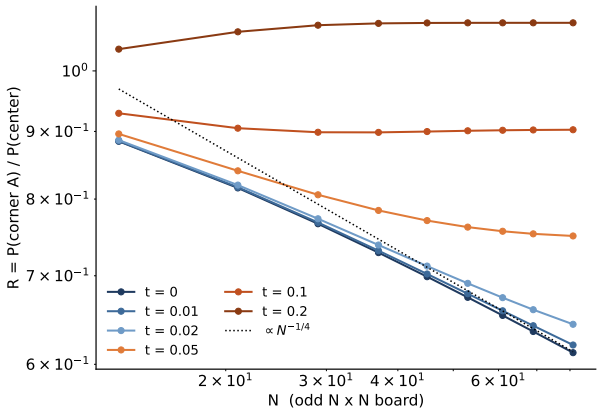


FIG. 7. Corner-to-centre ratio R of a forced defect on odd $N \times N$ boards ($N \equiv 1 \pmod 4$; the branch $N \equiv 3$ behaves alike). At $t = 0$, $R \propto N^{-1/4}$ (dotted); for $t > 0$, R saturates at $N \sim 1/t$.

closed form.

A. Boundary trapping stops at $N \sim 1/t$

On odd $N \times N$ boards one defect is still forced. Figure 7 shows the ratio of its probability at corner A to that at the centre. At $t = 0$ it decays as $N^{-1/4}$. For $t > 0$ it saturates once N exceeds roughly $1/t$ (for $t \le 0.02$ beyond the sizes shown), and the entropic force that held the defect away from the corner disappears. The saturated values depend on the corner angle (Table IV): lattice-scale boundary information survives the screening. At $t = 1/2$ the value at corner B is $\sqrt{3}$ to 2×10^{-13}, for which we have no explanation.

A naive scaling collapse, $R(N,t)/R(N,0) = G(Nt^{\nu})$, gives an apparent $\nu \approx 1.26$, whereas the fermion mass $\propto t$ implies $\nu = 1$ [2]. The discrepancy has a simple origin. Diagonal bonds join sites of the same colour, so once the defect is fixed the parity of the number of diagonal dimers is fixed, and $Z(-t) = \pm Z(t)$. Hence $\ln R(N,t) - \ln R(N,0) = a_2(N)t^2 + O(t^4)$. An empirical form, $a_2(N) = cN^2 + eN \ln N + f$, fits with $c \approx 0.015$ for both corners and $e/c \approx 7$–8 [Fig. 15(b)]. The leading term $(Nt)^2$ is the one expected for $\nu = 1$; the large subleading term produces the apparent exponent. Fitted on $29 \le N \le 81$, this form predicts $a_2(101)$ to 0.1% and 0.01% for the two corners, against 0.3–0.4% for a three-parameter power law $aN^b + f$.

B. Pair correlations and screening lengths

For monomers n sites apart on adjacent rows, Basor and Bleher proved $C(n) - C_\infty \simeq C_\infty e^{-2n \ln \eta_2}[C_1 + (-1)^{n+1}C_2]/(2n)$ for $0 < t < 1/2$, with $\eta_2^{-2} = 2 - \mu - 2\sqrt{1 - t^2 - \mu}$, $\mu = \sqrt{1 - 4t^2}$ [32]. Our exact values for monomers on the same row, up to $r = 80$ on 256×256 boards, show the same rate and the r^{-1} prefactor, with a dominant alternating part (Fig. 8); at $r = 80$ they agree with C_∞ [31] to 10^{-13} for $t \ge 0.15$. The screening is mildly anisotropic. Along the anti-diagonal $(1, -1)$, which no diagonal bond joins, the decay rate per unit Euclidean length, estimated from $|\Delta r| = 7$ and 9, is within about 15% of $2 \ln \eta_2$ for $t \le 0.4$ but is smaller

TABLE IV. Saturated corner-to-centre ratios ($N = 61$, converged to 10^{-12}). Corner A is touched by a diagonal bond and becomes a 120° corner at $t = 1$; corner B becomes a 60° corner.

t	corner A	corner B
0.2	1.087	1.172
0.5	1.350	1.732
1.0	1.526	2.587

at $t = 1/2$, 1.05 against 1.32 [Fig. 15(a)]. The longest screening length is therefore $1/(2\ln\eta_2) \simeq 1/(2t)$ up to a factor of order one, the scale at which the corner trapping saturates.

C. A pure exponential at $t = 1/2$

Basor and Bleher identified $t_c = 1/2$ as critical: there $\eta_1 = \eta_2$ and the asymptotic form changes from monotone to oscillating; their theorems exclude this point [32]. (The system is gapped for every $t > 0$.) At $t = 1/2$ we find, to 10^{-10} on finite boards,

$$C(\Delta r, \Delta c) = \frac{1}{6}\Big[1 - (\sqrt{3}-2)^{|\Delta r|+|\Delta c|}\Big] \qquad (\Delta r\,\Delta c \geq 0), \tag{8}$$

along $(1,0)$, $(0,1)$, $(1,1)$ and $(1,2)$ (Fig. 16). Along the anti-diagonal the decay is slower and not purely exponential. The rate $\ln(2+\sqrt{3})$ is the limit of $2\ln\eta_2$ and $C_\infty = 1/6$ is the Basor–Ehrhardt limit [31] evaluated at $t = 1/2$; what is new is that the correction has no power-law prefactor. The structure of the symbol is consistent with this. In the representation $C = \frac{1}{2}\sqrt{\det T_n(\phi)}$ of Ref. [32], ϕ contains $[t^2 + u + u^2]^{-1/2}$ with $u = \sin^2 x$. For $t < 1/2$ the two roots of $u^2 + u + t^2$ produce square-root branch points, whose Fourier coefficients decay as $k^{-1/2}\eta_2^{-k}$ and give the r^{-1} prefactor. At $t = 1/2$ the roots merge, $u^2 + u + \frac{1}{4} = (u + \frac{1}{2})^2$, the square root disappears, and the symbol becomes rational, for which $\det T_n$ is a finite sum of exponentials. Evaluating the block determinant with 40-digit arithmetic gives $C = [1 - (\sqrt{3}-2)^{n+1}]/6$ for $n = 1, \ldots, 30$ to 10^{-41}, and $4C^2$ satisfies a linear recurrence of order three. That C itself, rather than only C^2, is a two-term sum, and the amplitude $1/6$, are not explained by this argument; Eq. (8) is a conjecture verified to 40 digits. It also satisfies the exact local sum rule $\sum_j w_{ij} C(i,j) = 1$, which holds because site i is matched to exactly one neighbour: $4C(1,0) + 2tC(1,1) = \frac{2}{3}(3-\sqrt{3}) + \frac{1}{3}(2\sqrt{3}-3) = 1$ at $t = 1/2$. A correction without power-law prefactor at the point where the decay changes from monotone to oscillating is reminiscent of disorder points in Ising models. Near $t = 1/2$ the r^{-1} form holds only beyond a crossover length $\propto (1/2 - t)^{-1/2}$.

VII. DISCUSSION

On bipartite lattices, a forced defect goes where a charge e would go in the field of fractional corner charges, with closed edges acting as same-sign mirrors and the stiffness setting the overall power. Table V lists what was tested and what was found.

Two points deserve comment. First, the corner charge is fixed by the boundary height only up to the boundary layer. On the square lattice a mirror symmetry removes the layer's effect at every stiffness. On the honeycomb lattice it does not, and at 120° armchair corners the layer shifts the charge by $\pm\frac{1}{6}$, a shift that comes from the two edges meeting there rather than from the corner itself. Its size follows exactly from the Temperley property within the Coulomb-gas description; a direct lattice derivation, presumably a Wiener–Hopf factorization of the strip problem, is left open. Second, the binding criterion separates edges from the interior through the image alone. Whether the boundary-layer offsets of honeycomb edges change with interactions, which would make the corner charges there depend on g, is a question the present methods can answer.

ACKNOWLEDGMENTS

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Data availability. Codes and data are available at https://github.com/cywon-commits/Domino-Defects (branch `paper2`).

Appendix A: Exact methods

Kasteleyn and Pfaffian matrices. On the square and honeycomb lattices we use Kasteleyn signs [33–35] with every square or hexagonal face satisfying the clockwise-odd condition. A removed interior site creates a face that violates it; reversing the signs of the edges crossed by a string from the site to the boundary repairs it. For the diagonal model of Sec. VI, horizontal edges point east, vertical edges alternate up and down between columns, and each diagonal follows the vertical edge of the column to its right; every triangular face then has an odd number of clockwise edges (Fig. 9). All constructions were checked against brute-force enumeration: every defect site on $N \times N$ boards with $N = 5, 7, 9$, all pairs on 4×4 and 6×6 boards, every defect site of small honeycomb flakes, and 83 fixed-charge configurations on small square boards.

Fixed charges. On an $L \times L$ board with $m = L/2$, the interior $q = 1$ charge is the cell (m, m) and the edge one $(0, m)$. The interior $q = 3$ hole is the plus centred at $(m, m{+}1)$, with the balancing white cells at $(L/4, L/4{+}1)$ and $(3L/4, 3L/4{+}1)$; the edge hole is the plus centred at $(1, m)$, with white cells at $(m, L/4 + 1)$ and $(m, 3L/4 + 1)$. The balancing cells are fixed charges -1 at distances proportional to L.

Precision. Ratios of partition functions are computed as $\log|\det|$ of the string-modified matrix by sparse LU (SuperLU with COLAMD ordering), separately for every defect position. This avoids the loss of digits of Schur-complement updates when correlations are exponentially small and agrees with dense determinants to 10^{-11}. Edge

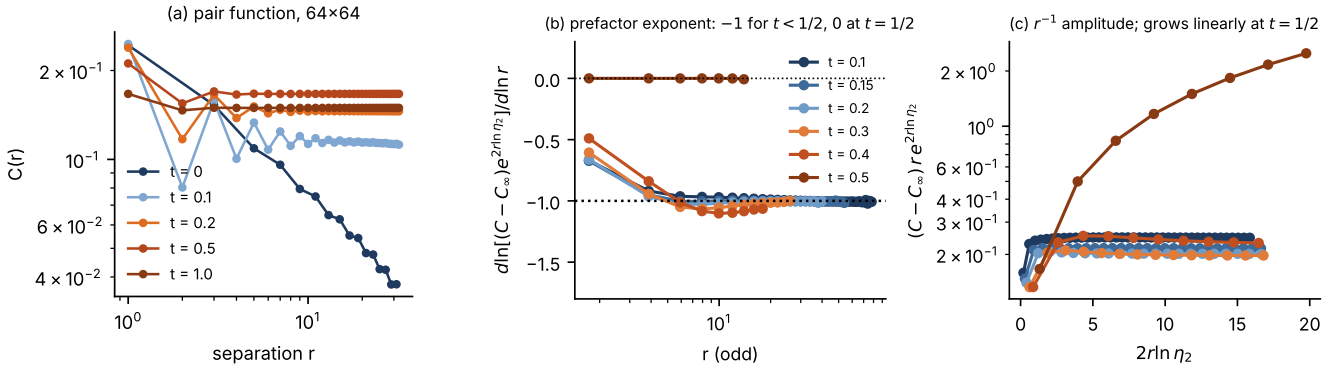


FIG. 8. (a) Bulk pair function $C(r)$ on 64×64 boards; at $t=0$ even separations vanish by the colour constraint. (b,c) Same-row data on 256×256 boards (384×384 at $t=0.1$), odd r. (b) Local prefactor exponent of $(C-C_\infty)e^{2r\ln\eta_2}$; it tends to -1 for $t<1/2$ and to 0 at $t=1/2$. (c) $(C-C_\infty)\,r\,e^{2r\ln\eta_2}$: flat for $t<1/2$, growing linearly at $t=1/2$ where the r^{-1} prefactor is absent.

TABLE V. Tests of the electrostatic picture.

ingredient	prediction	outcome
image doubling at edges	binding at an edge for $eq<-\frac{1}{g}-\frac{1}{4}$, in the interior for $eq<-\frac{2}{g}$	exact, $g=1/2$: $eq=-3$ bound at an edge ($P_\infty\approx0.34$), free in the interior; MC at $g=1.88$: for a single vacancy the local slope falls toward 0 at an edge ($-0.23\to-0.10$), not in the interior (-0.49)
stiffness	every exponent $\propto 2g$	edge/bulk pair ratio 2.01 ± 0.05 for $0.09\le g\le1.9$; corner exponent $=g/2$ within 4% for $g\le1$, 9% low at $g=1.88$
Temperleyan regions	$eq_c=-\frac{1}{2}(1-\theta_c/\pi)$, $\gamma=g/2$	type I hexagons and F2 parallelograms (boundary uniform to 10^{-13}); fixes $2\delta=\frac{1}{2}$
boundary layer	square: $\delta=0$ by symmetry; honeycomb: $\delta_-=-\delta_+$	honeycomb strip: $\delta_--\delta_+=0.5001\pm0.0010$; non-Temperleyan corners (type II, F1) agree within 1σ; full densities 3–5%
image sign	set by the boundary type	same sign on armchair edges, as for constrained boundaries [20]
bipartiteness	needed for the height	broken: trapping saturates at $N\sim1/t$ with angle-dependent limits; at $t=1/2$ a pure exponential, consistent with a rational symbol

probabilities are $P_{ab}=K_{ab}K_{ba}^{-1}$, and mean heights are integrated on the dual graph with curl errors below 4×10^{-15}.

Infinite armchair strip. Fourier transformation along the edge reduces the strip to a finite Kasteleyn matrix for each wave number. Bond probabilities are integrals over the wave number, converged by doubling the grid; the spread between grid sizes is listed with the data. The mean height across the strip follows by integration; its interior profile is linear, and its extrapolation to the two edges defines $\delta_\pm$.

Conformal map of the hexagon. The regular hexagon is the image of the unit disk under the Schwarz–Christoffel map $z=C\,e^{i\pi/6}\int_0^\zeta(1-s^6)^{-1/3}\mathrm{d}s$ [36], and the disk is mapped to the upper half plane by $f=i(1+u\zeta)/(1-u\zeta)$ with $u=e^{-i\pi/6}$. The map is inverted numerically with 8000 quadrature nodes. Lattice positions are scaled so that the apothem is R to $R+1$ lattice spacings.

Appendix B: Monte Carlo

Pair weights. On a cylinder the worm tail is fixed at a chosen site and the head performs directed-loop moves [17, 29] with the interaction weight in the acceptance ratio. The histogram of head positions is proportional to $Z(\text{tail},\text{head})$. The algorithm reproduces exhaustive enumeration on 4×4 and 6×4 cylinders to 3×10^{-4} for $\beta v=0,\pm1$.

Single forced defect. On an open board, with or without removed sites, two moves alternate with equal probability. A slide moves the defect from m to k by shifting the dimer (j,k) that covers a neighbour j of m to (m,j); it is accepted with probability $\min[1,(z_m/z_k)\,e^{-\beta v\Delta N_\parallel}]$, where z_m is the number of available neighbours of m. A

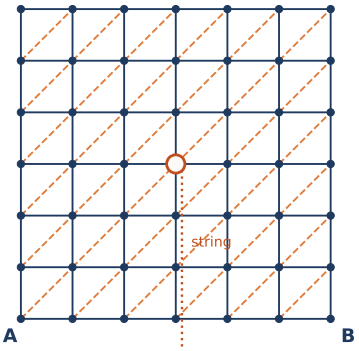


FIG. 9. Square lattice with diagonal bonds of weight t (dashed). A defect (open circle) is attached to the boundary by a string (dotted) along which edge signs are reversed. Corners A (touched by a diagonal) and B become 120° and 60° corners of the triangular lattice.

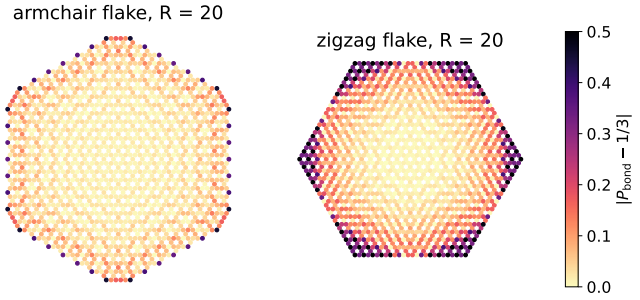


FIG. 10. Local deviation $|P-1/3|$ of the exact bond probabilities. Armchair flakes are flat liquids (core mean 0.035, 0.017, 0.012 for $R = 12, 20, 28$); zig-zag flakes (lozenge hexagons) freeze near the corners.

plaquette flip rotates two parallel dimers with Metropolis probability. Slides alone are not ergodic, because a vacancy confined by a fixed background explores only a finite domain [11]; the flips make the chain ergodic. The histograms agree with exhaustive enumeration on 5×5, 4×4 (one removed site) and 6×4 (one removed site) boards for $\beta v = \pm 1$, the largest deviation being 1.3 standard errors (4×10^8 moves). Production runs use $6 \times 10^5 L^2$ moves after 10% equilibration, split into ten blocks; errors are block standard errors and jackknife errors for fitted exponents.

Appendix C: Additional figures

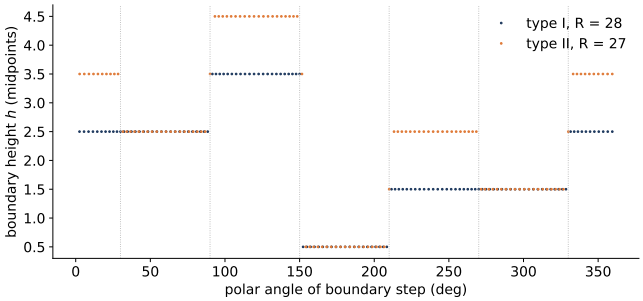


FIG. 11. Boundary height (step midpoints) around type I and type II hexagons with one forced defect. It is constant along each edge and jumps at the corners; the jump of -3 is the branch cut of the defect.

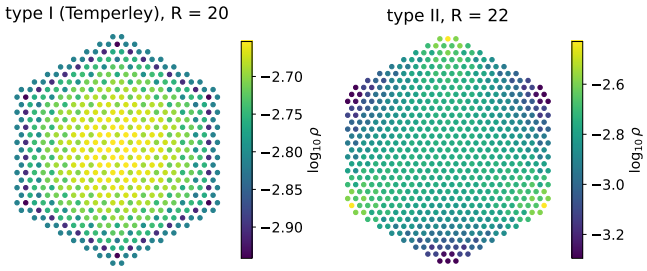


FIG. 12. Exact defect density on A sites of a type I (left) and a type II (right) hexagon (separate colour scales). The type I density is uniform along the boundary and sixfold symmetric.

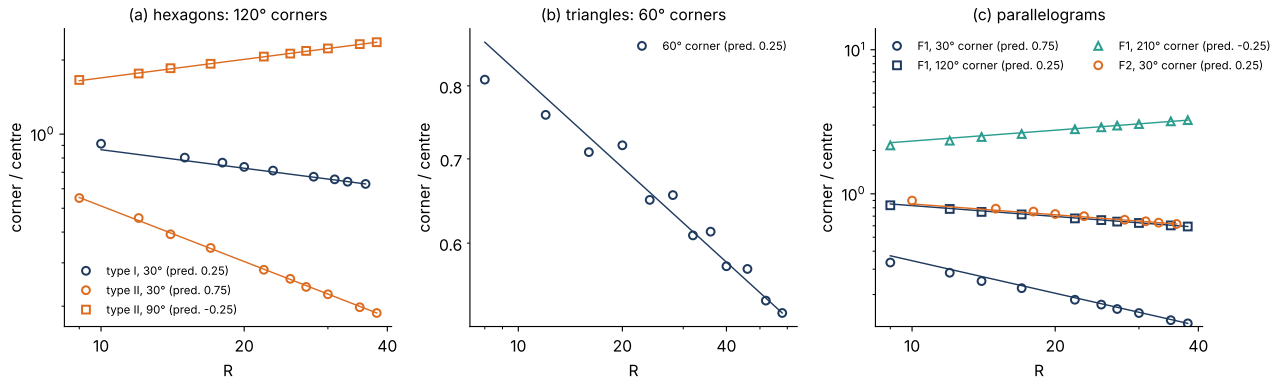


FIG. 13. Corner-to-centre ratios against size R; lines have the slopes predicted by Eq. (7), through the largest size. (a) Hexagons. (b) Triangles. (c) Parallelograms; the four F2 corners coincide, so one is shown.

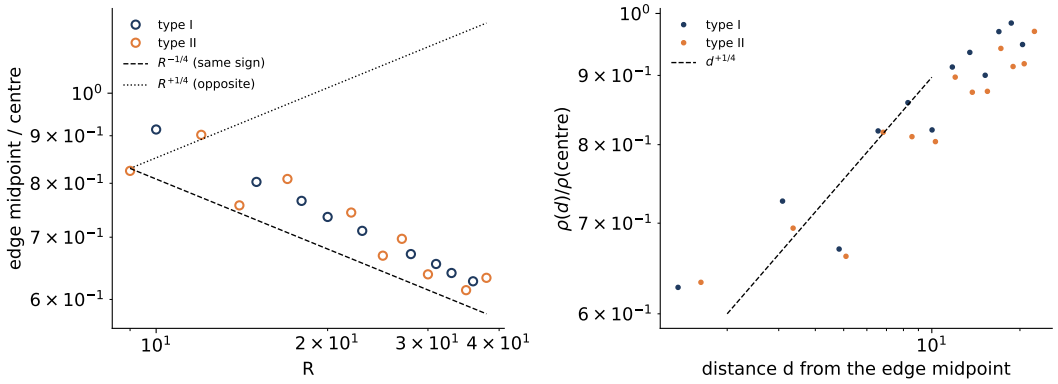


FIG. 14. Same-sign images on armchair edges. (a) Edge-midpoint to centre ratio against size; dashed: $R^{-1/4}$ (same sign), dotted: $R^{+1/4}$ (opposite sign). (b) Inward profile from the edge midpoint.

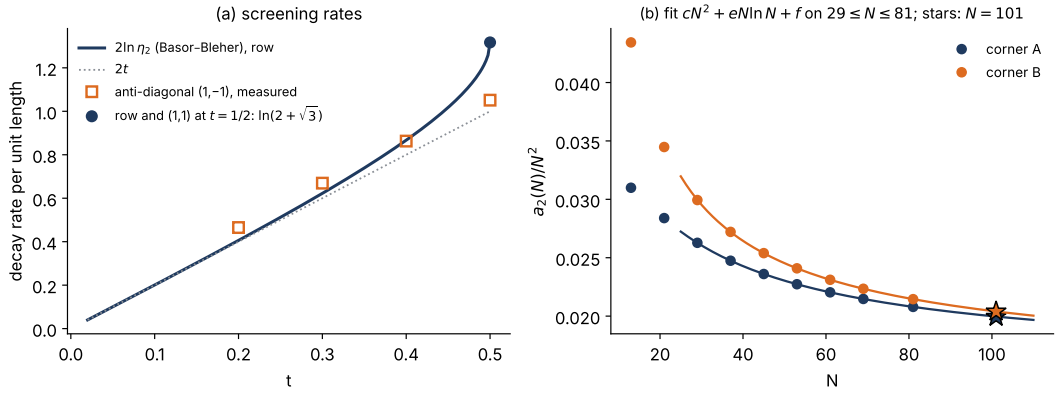


FIG. 15. Diagonal model. (a) Decay rate of $C - C_\infty$ per unit Euclidean length: $2\ln\eta_2$ of Basor and Bleher (line), measured along the anti-diagonal (squares, from $|\Delta r| = 7$ to 9), and $\ln(2+\sqrt{3})$ along rows and $(1,1)$ at $t = 1/2$ (circle). (b) $a_2(N)/N^2$ for both corners; lines are fits $cN^2 + eN\ln N + f$ on $29 \le N \le 81$, stars are $N = 101$ (not used in the fit).

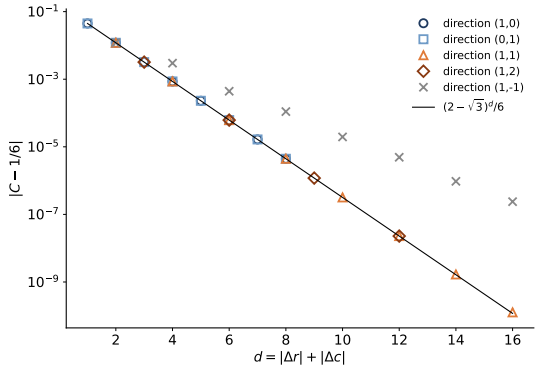


FIG. 16. $|C - 1/6|$ at $t = 1/2$ against the Manhattan distance $d = |\Delta r| + |\Delta c|$ for several directions; the line is $|\sqrt{3} - 2|^d/6$, Eq. (8). The anti-diagonal $(1, -1)$ decays more slowly.

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